HSF302 - JPA and Spring Boot Architecture Guide - Page 1

Chapter 1: Entity Lifecycle in Java Persistence API (JPA)

In JPA and Hibernate, an entity can exist in four primary lifecycle states: New/Transient, Managed/Persistent, Detached, and Removed.

1. New/Transient: The entity has just been instantiated using the Java new keyword and is not associated with any EntityManager persistence context. It has no primary key identifier in the database.

2. Managed/Persistent: The entity is associated with an active EntityManager session. Any modifications to its fields are tracked by Hibernate dirty checking mechanism and flushed to the database on commit.

3. Detached: The entity has a primary key representation in the database, but its EntityManager context was closed or cleared.

4. Removed: The entity has been marked for deletion via entityManager.remove() and will be deleted from the database table upon transaction commit.

HSF302 - JPA and Spring Boot Architecture Guide - Page 2

Chapter 2: Spring Data JPA Repositories and Query Generation

Spring Data JPA simplifies the data access layer by automatically implementing interface methods extending JpaRepository.

- Derived Query Methods: Method names like findByUsernameIgnoreCase or findAllByCourseIdOrderByUploadedAtDesc are parsed into SQL queries at application startup.
- Custom JPQL: The @Query annotation allows declaring custom queries against JPA entities rather than underlying database tables.
- Transaction Management: Service methods annotated with @Transactional ensure atomic operations with automatic rollback on unhandled RuntimeException.